

Solar O appendix

Sun · O · downloadable appendix

Solar oxygen: 8.660 ± 0.101

Asplund, Amarsi & Grevesse 2021: 8.69 ± 0.04 · this measurement -0.030 dex

Kitt Peak (Kurucz 2005) · red-optical · Reference Grade · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · $n = 6$. The top-ranked admitted product (Reference Grade, then the most complete treatment, then line count, then total uncertainty); independent products and engines are not averaged together.

Highlighted Products

VIS

8.934 ± 0.393

± 0.309 stat · ± 0.243 syst

[O I] 6300 Å forbidden line (blended with Ni I) · 1D LTE · HARPS · Codex Grade

red-optical

8.660 ± 0.101

± 0.044 stat · ± 0.092 syst

O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (Kurucz 2005) · Reference Grade

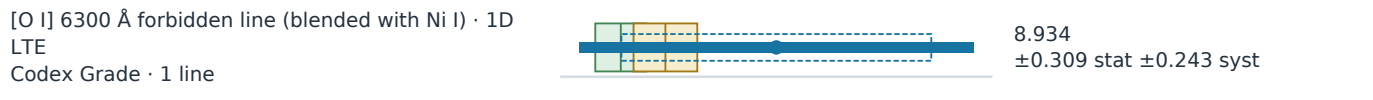
Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	9 products -- O I · Asplund et al. 2021 lines
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	10 products -- O I · Codex-graded lines; [O I] 6300 Å forbidden line (blended with Ni I)
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines)

Error-bar Forest Plot

VIS — HARPS A(O) dex; axis 8.60-9.27



VIS axis: 8.60 · 8.76 · 8.93 · 9.10 · 9.27 A(O) dex

red-optical — IAG FTS A(O) dex; axis 8.52-8.99



red-optical — IAG FTS A(O) dex; axis 8.52-8.99

O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)
Reference Grade · 6 lines



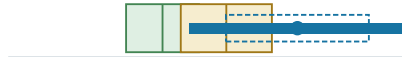
8.659
 ± 0.045 stat ± 0.097 syst

O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)
Reference Grade · 6 lines



8.689
 ± 0.040 stat ± 0.094 syst

O I · Codex-graded lines · 1D LTE
Codex Grade · 2 lines



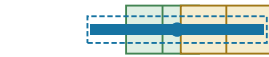
8.838
 ± 0.119 stat ± 0.078 syst

O I · Codex-graded lines · 3D NLTE (Amarsi)
Codex Grade · 2 lines



8.677
 ± 0.108 stat ± 0.102 syst

O I · Codex-graded lines · 1D NLTE (Bergemann)
Codex Grade · 2 lines



8.706
 ± 0.095 stat ± 0.098 syst

red-optical — Kitt Peak (Kurucz 2005) A(O) dex; axis 8.52-8.99

O I · Asplund et al. 2021 lines · 1D LTE
Reference Grade · 6 lines



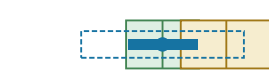
8.820
 ± 0.048 stat ± 0.065 syst

O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)
Reference Grade · 6 lines



8.660
 ± 0.044 stat ± 0.092 syst

O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)
Reference Grade · 6 lines



8.690
 ± 0.038 stat ± 0.089 syst

O I · Codex-graded lines · 1D LTE
Codex Grade · 2 lines



8.841
 ± 0.122 stat ± 0.078 syst

O I · Codex-graded lines · 3D NLTE (Amarsi)
Codex Grade · 2 lines



8.680
 ± 0.111 stat ± 0.100 syst

O I · Codex-graded lines · 1D NLTE (Bergemann)
Codex Grade · 2 lines



8.709
 ± 0.098 stat ± 0.098 syst

red-optical — Kitt Peak (1984, molecfit) A(O) dex; axis 8.52-8.99

O I · Asplund et al. 2021 lines · 1D LTE
Reference Grade · 6 lines



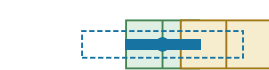
8.820
 ± 0.052 stat ± 0.064 syst

O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)
Reference Grade · 6 lines



8.660
 ± 0.047 stat ± 0.090 syst

O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)
Reference Grade · 6 lines



8.690
 ± 0.041 stat ± 0.088 syst

O I · Codex-graded lines · 1D LTE
Codex Grade · 2 lines



8.833
 ± 0.132 stat ± 0.077 syst

O I · Codex-graded lines · 3D NLTE (Amarsi)
Codex Grade · 2 lines



8.672
 ± 0.121 stat ± 0.100 syst

O I · Codex-graded lines · 1D NLTE (Bergemann)
Codex Grade · 2 lines



8.701
 ± 0.107 stat ± 0.097 syst

red-optical axis: 8.52 · 8.64 · 8.76 · 8.88 · 8.99 A(O) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every oxygen value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the oxygen abundance as the free parameter. The synthesis includes the atomic lines and the molecular bands (CN, CH, C₂, OH and others) in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, blends and coupling to the other C/N/O abundances, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	What was measured	Solar spectra	Treatments	Lines	Result A(O)	Largest uncertainty terms (dex)
VIS	[O I] 6300 Å forbidden line (blended with Ni I)	HARPS	1D-LTE	1	8.934 (1D-LTE)	fit / line-to-line scatter 0.31, continuum placement 0.17
red-optical	O I · Codex-graded lines	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	2	8.672–8.680 (3D-NLTE)	fit / line-to-line scatter 0.11, NLTE / 3D correction 0.07
red-optical	O I · Asplund et al. 2021 lines	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	6	8.659–8.660 (3D-NLTE)	NLTE / 3D correction 0.07, fit / line-to-line scatter 0.04

Solar oxygen context

Oxygen is traced by forbidden [O I], permitted O I and molecular OH indicators where available. Blends, departures from LTE and atmospheric structure affect these families differently, so their results retain separate identities.

[C/O indicator strategy](#) · [Methodology](#)

Open provenance decisions

O I 615.8nm: □ DISAGREES — reported to Ryan, not resolved. . Store log gf -0.409 (); summed pull -0.2966 (B+).

O I 630.0nm: □ DISAGREES — reported to Ryan, not resolved. □ store claims A, pull publishes B+. Store log gf -9.717 (A); summed pull -9.7744 (B+).

OPEN. Tachiev and NIST differ systematically by -0.016 dex on the 777 triplet and -0.005 dex on 8446. AGSS21 names Tachiev for N i only and states no O i gf source. Both values are reported in oi_tachiev_vs_nist.csv; NIST is what the store already carried and this script does not change it. Ryan decides.

[Adjudication record](#) · [Indicator gf checks](#)

Download the evidence

[Download PDF — Solar O appendix](#)

[Source feed — includes quarantined fit outputs, not adopted abundances](#)

[Complete product visibility audit](#)

[Shared website/PDF report](#)

[Reproducibility manifest](#)

[Complete references and product-to-source bindings](#)

Reproducibility

Generated from the pinned science snapshot. Source commit c730d5e72e3f8902d1d4d800b46fe89fbcafee69

Feed O.json v1.74 · updated 2026-10-03T22:03:37Z

Generated 2026-10-09T16:35:54-06:00 (source commit timestamp)

Feed SHA-256 bb076290a638dc0cdf74cf42cc9a553e2af94f140fa3c415c9ee20da4fed99a7

Generator scripts/generate_cno_publication.py v1.0.0

[Pinned scientific source](#)

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar O atomic/molecular indicator lineage and atmospheric sensitivity; comparison literature, not an adopted Codex abundance. Sources: [1], [2].

Atomic data and gf authorities. The recorded O I/O II base pedigree is Wiese-Fuhr-Deters / Opacity Project theory. Per-transition exceptions and the open O I source decisions remain explicit. Sources: [3], [4], [5].

Atomic data and gf authorities. O I triplet source record: NIST — Tachiev NOT adopted; OPEN — under review Sources: [3], [6].

Model and grid sources. Registered O I line/grid lineage: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — line-by-line 3D-NLTE grids built on the Amarsi+2018 O I model atom. The current feed treatment remains authoritative; a grid reference does not imply a correction was applied. Sources: [7], [8].

Atomic data and gf authorities. Atomic O I entry; source-family pedigree only, not a substitute for a per-transition gf join. Sources: [3], [4], [5].

Model and grid sources. Feed treatment: 1D-LTE; no NLTE/3D correction is claimed for this entry. Turbospectrum is the synthesis implementation. Sources: [9].

Special diagnostics. The synthesis sidecar names Johansson's Ni I blend data, Storey & Zeippen's forbidden-line gf lineage and Caffau's CO5BOLD comparison anchor. The feed entry remains 1D-LTE and quarantined; the sidecar's cited anchor is not promoted to a measured abundance. Sources: [10], [11], [12].

Model and grid sources. Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 3D non-LTE - 1D LTE on STAGGER (table2 C I / table3 O I); PER-LINE additive correction, grid lines only (match <= 0.05 A) Sources: [7], [8].

Model and grid sources. Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 1D non-LTE - 1D LTE (table5 C I / table6 O I); PER-LINE additive correction, grid lines only (match <= 0.05 A) Sources: [7], [8].

Product-to-source index

VIS · HARPS · [O I] 6300 Å forbidden line (blended with Ni I) · 1D LTE · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#), [\[10\]](#), [\[11\]](#), [\[12\]](#)

O-products-0ba7067733f9

red-optical · IAG FTS · O I · Asplund et al. 2021 lines · 1D LTE · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#) O-products-d00ae41a9b00

red-optical · IAG FTS · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-a911fb18ea46

red-optical · IAG FTS · O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-bbe6b69b508ea

red-optical · IAG FTS · O I · Codex-graded lines · 1D LTE · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#) O-products-c4bdfbe8ac19

red-optical · IAG FTS · O I · Codex-graded lines · 3D NLTE (Amarsi) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-f02b3eb1b0e5

red-optical · IAG FTS · O I · Codex-graded lines · 1D NLTE (Bergemann) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-1a72432a141c

red-optical · Kitt Peak (Kurucz 2005) · O I · Asplund et al. 2021 lines · 1D LTE · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

O-products-711c33d498f6

red-optical · Kitt Peak (Kurucz 2005) · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-297d2f9df123

red-optical · Kitt Peak (Kurucz 2005) · O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#) O-products-24673d0fe3f0

red-optical · Kitt Peak (Kurucz 2005) · O I · Codex-graded lines · 1D LTE · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

O-products-4961ac96e190

red-optical · Kitt Peak (Kurucz 2005) · O I · Codex-graded lines · 3D NLTE (Amarsi) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-e0acb098043b

red-optical · Kitt Peak (Kurucz 2005) · O I · Codex-graded lines · 1D NLTE (Bergemann) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-294d60b5a66b

red-optical · Kitt Peak (1984, molecfit) · O I · Asplund et al. 2021 lines · 1D LTE · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

O-products-1d0c52443ad4

red-optical · Kitt Peak (1984, molecfit) · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#) O-products-84774ef30efe

red-optical · Kitt Peak (1984, molecfit) · O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#) O-products-faf80741533b

red-optical · Kitt Peak (1984, molecfit) · O I · Codex-graded lines · 1D LTE · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

O-products-e7b874120b2b

red-optical · Kitt Peak (1984, molecfit) · O I · Codex-graded lines · 3D NLTE (Amarsi) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-093a3dc3a2d6

red-optical · Kitt Peak (1984, molecfit) · O I · Codex-graded lines · 1D NLTE (Bergemann) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#) O-products-b0c45bc56d31

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. *A&A* **653**, A141 DOI: 10.1051/0004-6361/202140445. [source record](#)

[2] — Amarsi et al. 2021, *A&A* **656** A113 DOI: 10.1051/0004-6361/202141384. [source record](#)

Atomic data and gf authorities

[3] — Kramida, A.; Ralchenko, Yu.; Reader, J.; NIST ASD Team (2024). NIST Atomic Spectra Database (version 5.11/5.12). National Institute of Standards and Technology, Gaithersburg, MD DOI: 10.18434/T4W30F. source record

[4] — Wiese, W. L.; Fuhr, J. R. (2006). New Critical Compilations of Atomic Transition Probabilities for Neutral and Singly Ionized Carbon, Nitrogen, and Iron. Proceedings of the NASA Laboratory Astrophysics Workshop (14-16 February 2006, UNLV, Las Vegas) source record

[5] — Wiese, W. L.; Fuhr, J. R.; Deters, T. M. (1996). Atomic Transition Probabilities of Carbon, Nitrogen, and Oxygen: A Critical Data Compilation. JPCRD Monograph 7. source record

[6] — Tachiev & Froese Fischer 2002, A&A 385, 716 DOI: 10.1051/0004-6361/20011816. source record

Model and grid sources

[7] — Amarsi, A. M.; Nissen, P. E.; Skuladottir, A. (2019). Carbon, oxygen, and iron abundances in disk and halo stars. A&A 630, A104 DOI: 10.1051/0004-6361/201936265. source record

[8] — Amarsi, Barklem, Asplund, Collet & Zatsarinny 2018, A&A 616, A89 DOI: 10.1051/0004-6361/201832770. source record

[9] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. A&A 669, A43 DOI: 10.1051/0004-6361/202243673. source record

Special diagnostics

[10] — Johansson, S.; Litzen, U.; Lundberg, H.; Zhang, Z. (2003). Experimental f-Value and Isotopic Structure for the Ni I Line Blended with [O I] at 6300 Å. ApJ 584, L107 DOI: 10.1086/374037. source record

[11] — Storey, P. J.; Zeippen, C. J. (2000). Theoretical values for the [O III] 5007/4959 line-intensity ratio and homologous cases. MNRAS 312, 813 DOI: 10.1046/j.1365-8711.2000.03184.x. source record

[12] — Caffau, E.; Ludwig, H.-G.; Steffen, M.; Livingston, W.; Bonifacio, P.; Malherbe, J.-M.; Doerr, H.-P.; Schmidt, W. (2015). The photospheric solar oxygen project. III. Investigation of the centre-to-limb variation of the 630 nm [O I]-Ni I blend. A&A 579, A88. source record